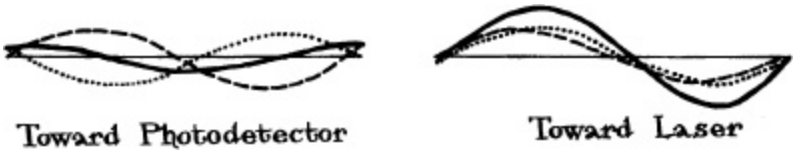


Toward the photodetector, the waves interfere perfectly destructively, so the total, superimposed wave vanishes, which means that the photodetector sees no light at all. When a gravitational wave or other force has lengthened op.e arm slightly and shortened the other, then the beam from the one arm arrives at the beam splitter with a slight delay relative to the other, and the superimposed waves therefore look like this:



The destructive interference in the photodetector's direction is no longer perfect; the photodetector receives some light. The amount it receives is proportional to the arm length difference, $L_1 - L_2$ which in turn is proportional to the gravitational-wave signal.

It is interesting to compare a bar detector with an interferometer. The bar detector uses the vibrations of a single, solid cylinder to monitor the tidal forces of a gravitational wave. The interferometric detector uses the relative motions of masses hung from wires to monitor the tidal forces.

The bar detector uses an electrical sensor (for example, piezoelectric crystals squeezed by the bar) to monitor the bar's wave-induced vibrations. The interferometric detector uses interfering light beams to monitor its masses' wave-induced motions.

The bar responds sympathetically only to gravitational waves over a narrow frequency band, and therefore, decoding the waves' symphony would require a xylophone of many bars. The interferometer's masses wiggle back and forth in response to waves of *all* frequencies higher than about one cycle per second,[7] and tllerefore the interferometer has a wide bandwidth; three or four interferometers are sufficient to fully decode the symphony.